

Takeru J. Hiura

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PROFESSIONAL EXPERIENCE

- Photonic Test Engineer Co-op** | Nokia Bell Labs | Murray Hill, New Jersey *Aug. 2026 - Present*
- Build and maintain DC power filter test benches using Chroma three-phase power supplies, programmable load boards, and ADC measurements for automated factory power-load testing.
 - Write Python scripts and test cases to automate hardware verification and measurement collection.
 - Debug and characterize hardware using oscilloscopes, optical, RF, and electrical test equipment.
- Data Analyst Intern** | Hyphenova | Remote *Aug. 2026 - Present*
- Create and test data pipelines and application workflows, including social-media scraping, platform bring-up, and validation of collected data and system outputs.
 - Support database implementation, schema design, data cleanup, and ERD documentation.
- Hardware Development Co-op** | Nokia Bell Labs | Murray Hill, New Jersey *Jan. 2026 - Aug. 2026*
- Developed FPGA verification and debug infrastructure using Verilog/VHDL, Python automation, and embedded logic analyzers to validate prototype evaluation boards and accelerate signal-level debugging.
 - Tested **100+** optical/telecom modules across **20+** validation procedures covering thermal characterization, signal integrity, power behavior, ripple, and insertion loss.
 - Performed board bring-up, CPU/DRAM stress testing, firmware integration, and interface debugging using oscilloscopes, logic analyzers, protocol analyzers, and other bench instrumentation.
- Group Tutor - Introduction to Computer Engineering** | Lehigh University | Bethlehem, PA *Aug. 2025 - Dec. 2025*
- Tutored **80+** students in digital logic, Boolean algebra, sequential circuits, and microcomputer architecture.
 - Guided students through digital circuit design, analysis, and assembly-level programming.

PROJECT EXPERIENCE

- FPGA-Based Real-Time Image Processing for License Plates** | [GitHub](#) *Aug. 2025 - Present*
- Design a Verilog FPGA video pipeline for **640×480** input at **30 FPS**, implementing RGB-to-grayscale conversion, 3×3 Sobel filtering, and OV7670 camera interfacing.
 - Optimize line buffering and memory architecture to keep BRAM utilization below **30%** while achieving **93%+** edge-detection accuracy in real-time processing.
 - Integrate FPGA-based image preprocessing with FastALPR for license-plate detection and OCR.
- Undergraduate Researcher** | The GP3 Lab | [Lab Website](#) *Feb. 2025 - Dec. 2025*
- Automated the processing and visualization of NetCDF climate datasets using Python, Matplotlib, and cmocean to generate high-resolution global dust distribution maps.
 - Benchmarked and tuned several spatial interpolation methods, reducing map generation time by lowering spatial resolution by **9x** while evaluating interpolation methods.
- Fresh from the Fridge** | AWS Hackathon Winner | [GitHub](#) *Oct. 2025*
- Implemented an end-to-end smart fridge system incorporating an OpenMV camera with AWS Lambda, S3, Textract, Rekognition, Bedrock, and Aurora DB to automatically log grocery items from video and receipt uploads.
 - Programmed embedded control logic in Python with a limit switch to trigger video capture, automating upload workflows and enabling cloud-based recognition and database updates.
- TM4C123 Embedded Smart Lock Prototype** | [GitHub](#) *Apr. 2025 - May 2025*
- Designed a TM4C123 smart lock with keypad/ADC authentication, PWM servo actuation, and Moore FSM control.
 - Implemented interrupt-driven IR intrusion detection, UART feedback, and watchdog-based input timeout with interrupt-driven control and subsystem-level validation.

EDUCATION

Lehigh University | Bethlehem, Pennsylvania *Expected Graduation: Dec. 2026*
Bachelor of Science in Computer Engineering *GPA: 3.98/4.00*
Related Coursework: Computer Architecture, FPGA Laboratory, Microcontroller Laboratory, Power Electronics
Honors: Upsilon Pi Epsilon, Tau Beta Pi, The Gabuzda Award, Dennis Ritchie Award, Eta Kappa Nu Prize, Dean's List

TECHNICAL SKILLS AND LANGUAGE PROFICIENCY

Programming & HDL: Verilog, VHDL, SystemVerilog, Python, C/C++, Bash, Java, Tcl, SQL, JavaScript, MIPS Assembly
Embedded Systems: FPGA development, microcontrollers, UART, I²C, SPI, PWM, ADC, interrupts, watchdog timers
Tools & Hardware: Xilinx Vivado, Lattice Diamond, PADS Professional, ODB++, Git, Linux, LTspice, MATLAB, AWS
Languages: English (Native), Japanese (Native)